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461L Building International Applications

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Localizing your App

1. Create a version in each language
2. Use Overlays
3. Create one version that detects the language of the device (or allows the user to choose) and switches resources accordingly



Option 1: A version for each Language

- Easiest to distribute
- Easiest to maintain
- Works on OS versions previous to 3.5



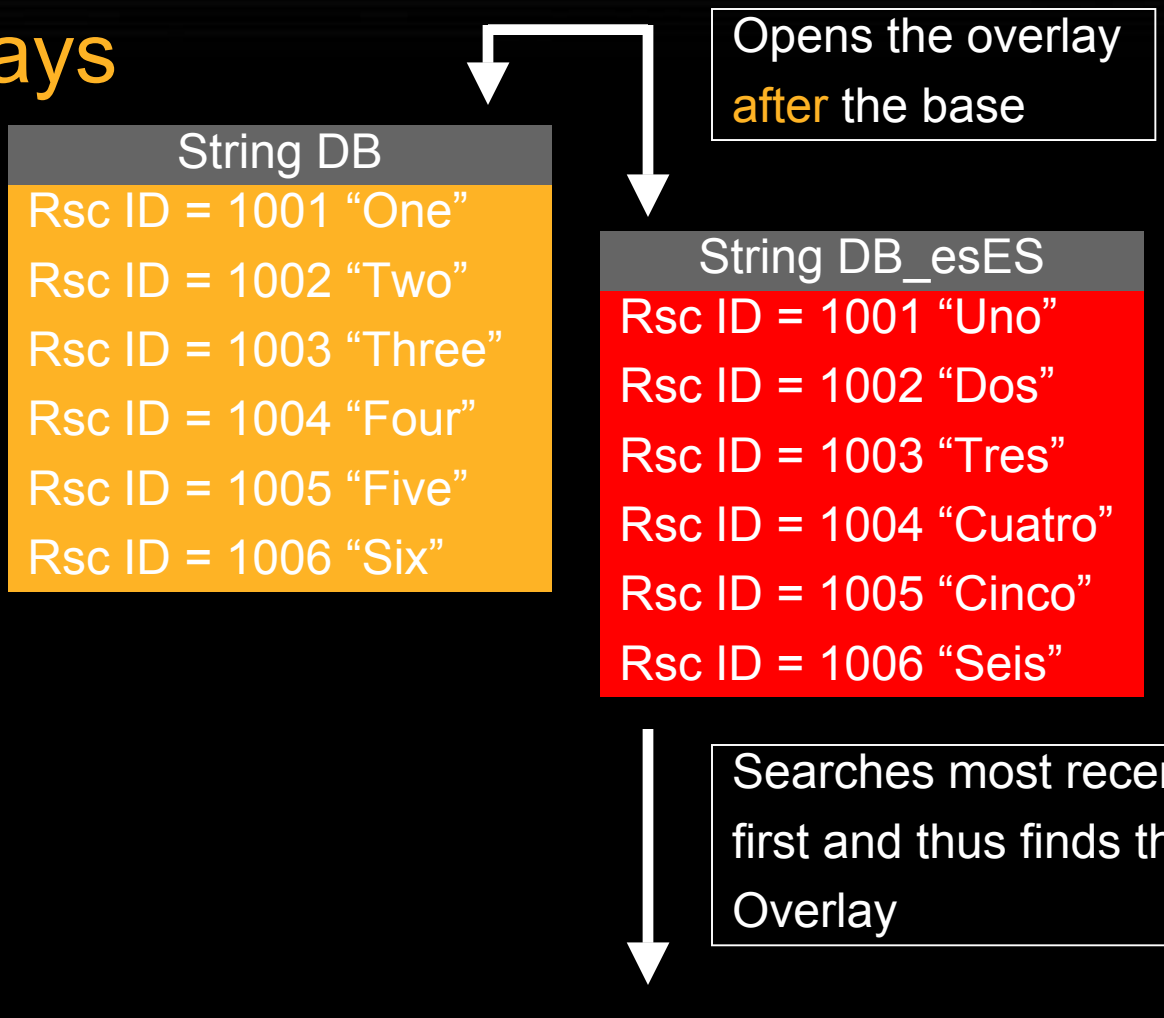
Demo 1: Setting CW with language targets

- Create esES and enUS subdirectories
- Duplicate resource files
- Localize esES
- Create esES Target
- Add esES resources to esES target

Option 2: Overlays

- Separate Localized PRC
- One Base PRC can have many Overlay PRCs
- Overlay Manager automatically opens overlay PRC and substitutes localized overlay resources at runtime
  - You usually do not call the Overlay Manager
  - If overlay PRC for current locale exists, the Overlay Mgr will open the overlay after the base db
  - You can specify not to use overlays, or call the Overlay Mgr to force the use of an overlay that does not match the current locale

Overlays

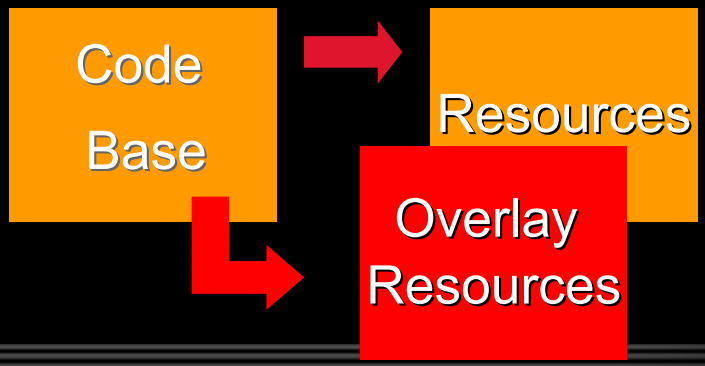



```
DmGetResource(StringRsc, 1003);
```

Overlays

- Name_InCN (example Name_enUS)
  - Use PRC2OVL -showlocales for list
- Overlay PRC type is ‘ovly’
- Overlay PRC creator is same as Base PRC
- ‘ovly’ =1000 that matches Base DB
- Matching Resource Ids, sizes and checksums







Demo 2: Look at Overlay with RsrcEdit

- Open POSE with EFIGS rom
- Add RsrcEdit.prc, myTstOverlay.prc and myTstOverlay_esES.prc
- Run RsrcEdit and look at overlays

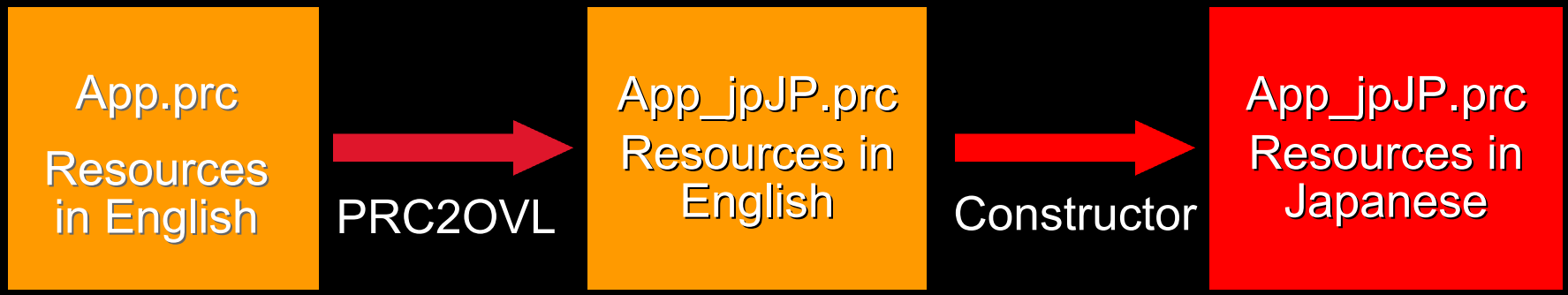
Making an Overlay

Method 1

- Start with Base PRC (any language)
- Create Overlay PRC with PRC2OVL Tool
- Edit resources in overlay with OnBoard RsrcEdit, PAR, PRC Constructor, etc.
- This will insure that the overlay will match up with the base
- You have to recreate the overlay if you alter non-code resources in base PRC

Making an Overlay

Method 1



Demo 3: Overlaying tstOverlay.prc

1. Open 461L Folder
2. Open PRC2OVL prompt
3. Type
   `"PRC2OVL tstOverlay.prc -locale jpJP -o tstOverlay_jpJP.prc"`
4. Open tstOverlay_esES.prc with Constructor
5. Change "This is English" to "This is Japanese"
6. Run POSE with Japanese rom
7. Load prc -> show, load overlay -> show again
8. Show same on German rom

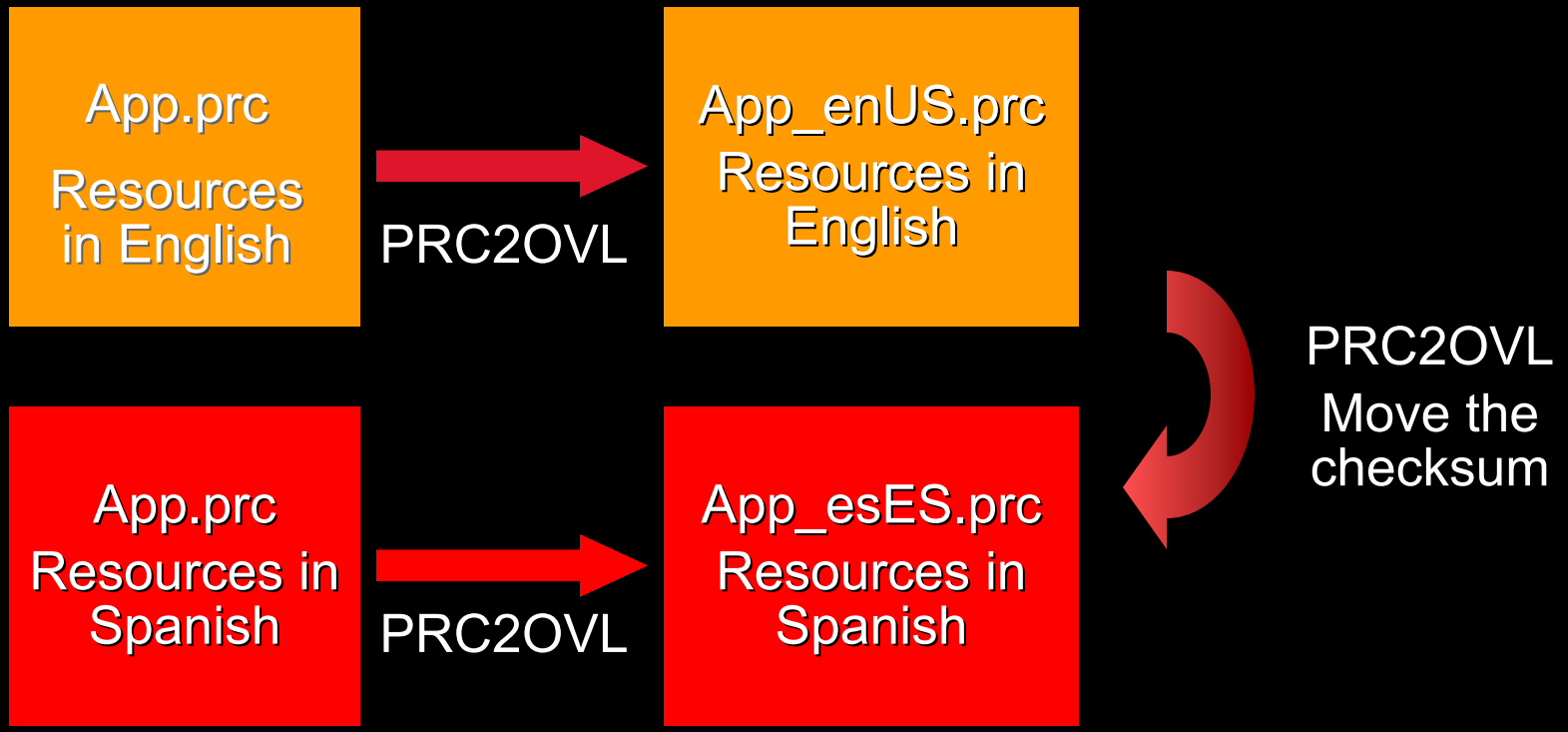
Making an Overlay

Method 2

- Create a Base Language PRC for each Language
- Choose 1 to be the shipping Base
- Create an Overlay from both PRCs
- Use the PRC2OVL tool's checksum modification to make them work (specifics)
- This makes it easier to change the base PRC and only that one part of the overlay
- You have to be careful to make sure that the UI matches on both PRCs!

Making an Overlay

Method 2





Demo 4: Create overlay from Spanish app

- Create overlay from Spanish app made above
- Create overlay from English app
- Drop checksum into Spanish overlay

Demo 5: (optional) modify and patch overlay

1. Modify tstOverlay.prc to have another label
2. Modify tstOverlay_esES.prc to have matching localized label
3. Load both on POSE with Spanish rom
4. Open PRC2OVL prompt
5. PRC2OVL tstOverlay.prc –c -p



You will now go read

Palm OS® Programming Developer Tools Guide

Part IV: Creating National Language Versions of your Application

Resources with Japanese

- You can
  - Use a Japanese system (or Japanese enabled Mac)
  - Enter hex values into constructor
  - In Win 2K, in regional settings enable Japanese fonts, input method AND set your default language to Japanese
  - Use a tool like RsrcEdit on POSE running a Japanese ROM



Demo: Entering Japanese

- Show how to set up Win 2k
- Show RsrcEdit

Delivering Overlays to the Handheld

- Get country and language and install proper overlay
  - `SyncReadFeature('psys', 5 /*country*/, &result);`
  - `SyncReadFeature('psys', 6 /*language*/, &result);`
  - See PalmLocale.h to decipher result
- Bind overlays into application
  - Create Overlay
  - Add overlay file as custom resource
  - Expand overlay on launch and relaunch
  - Come to 461lab for example…
- Trust the user to install the right one



Demo: Bind in Spanish Overlay

- Show binding of Spanish overlay
- Copy needed code from text file

Tool Links

- PRC2OVL
  - Comes with “Other Tools” from SDK
- OnBoard RsrcEdit
  - http://www.individeo.net/
- PRC Constructor
  - Will be in the new SDK :)
- PAR
  - http://www.djw.org/product/palm/par/index.html
- Multilizer
  - http://www.multilizer.com/solutions/tool/palm/
- ASCII Chart
  - http://www://astro.uchicago.edu/home/web/valdes/pilot/
- PowerGlot
  - http://www.powerglot.com



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